NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 20/01/2026
Time: 3 Hrs | GTM – 20 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	D	C	D	B	A	C	D	A	C
11	12	13	14	15	16	17	18	19	20
C	C	B	C	A	D	B	D	C	A
21	22	23	24	25					
3	2	13	5	4					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	C	D	D	C	A	A	B	D
36	37	38	39	40	41	42	43	44	45
B	D	A	D	C	D	B	A	B	A
46	47	48	49	50					
4	3	229	225	2					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	B	B	A	B	D	D	D	A	B
61	62	63	64	65	66	67	68	69	70
B	A	D	C	B	D	A	A	B	A
71	72	73	74	75					
5	145	72	6	3					

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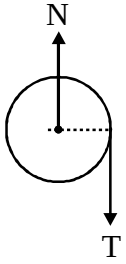
HINTS
PHYSICS

1. $E_0 = cB_0$

$\hat{E} = \hat{B} \times \hat{c}$

02. a/b has dimensions of energy.

03. Consider the angular momentum about O, we have



$Mv_0R = MVR + Mk^2\omega, v_0 = \sqrt{2gh}$

But $\omega = \frac{V}{R}$

$\therefore V = \frac{v_0}{\left(1+\frac{K^2}{R^2}\right)}$.

04. $\mu = -30cm$

$v = ?$

$f = -20cm \ \frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow v = -60$

$|m| = \left|\frac{v}{u}\right| = |-2| = 2$

$v_{sx} = \sqrt{5}\cos\left(\tan^{-1}2\right) = 1m/s = \hat{i}$

$v_{sx} = \sqrt{5}\sin\left(\tan^{-1}2\right) = 2m/s = +2\hat{j}$

$\frac{v_{lx}}{v_{ax}} = -m^2 = -4 \Rightarrow v_{tx} = -4m/s = -4\hat{i}$

$\frac{v_{ty}}{v_{oy}} m = -2 \Rightarrow v_{ly} = -4m/s = -4\hat{j}$

$v_t = \sqrt{v_{1y}^2 + v_{1x}^2} = 4\sqrt{2}m/s$

$v_{1/0} = \left(-5\hat{i} - 6\hat{j}\right)m/s,\ speed = \sqrt{5^2+6^2} = \sqrt{61}\,m/s$

05. $xa = yb$

$x + y = Q$

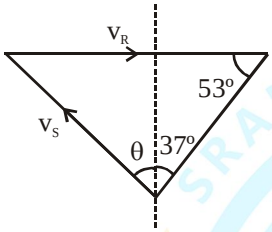
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$$F = \frac{xq}{A\epsilon_0}$$

06. $\tan\theta = \frac{45}{60} \Rightarrow \theta = 37°$

since, $V_S = V_R$,

$53° = \theta + 37°$, $\theta = 16°$.



07. $x = \frac{d}{1 \pm \sqrt{\frac{q_2}{q_1}}}$

08. $E = \frac{10 \times 50 \times 10 \times 30 \times 3600}{3.6 \times 10^6}$

09. x = 3 sin 2t + 4cos 2t

= 5 sin (2t + □/4)

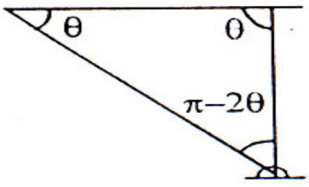
So A = 5 & $v_{max.} = A\square = 5 \times 2 = 10$

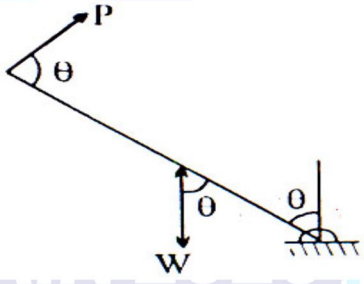
10. Conceptual

11. $I_z = \frac{7}{500} - \frac{5}{1000}$ A

12. Conceptual

13. $P\sin\theta = W\sin\theta\frac{l}{2}$





$$P = \frac{W\sin 2\theta}{2\sin\theta}$$

14. $B_X = \frac{\mu_0 I r^2}{2\left[r^2 + \left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}}$

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$$B_Y = \frac{\mu_0 I 4r^2}{2\left[4r^2 + d^2\right]^{\frac{3}{2}}}$$

15. Conceptual

16. $$I = 3\left[\frac{\left(\frac{m}{3}\right)\left(\frac{l}{3}\right)^2}{12} + \frac{m}{3}\left(\frac{1}{3}\left(\frac{\sqrt{3}l}{6}\right)^2\right)\right]$$

17. $$\frac{h^1}{h} = \frac{\mu_1 v}{\mu_2 u}$$

18. Conceptual

19. $$\text{Tan}\,\theta = \frac{\sigma Q}{2 \in_0 \in_r mg}$$

20. $E_3 - E_2 = 13.6\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$

$$= \frac{13.6 \times 5}{36} = 1.89 \text{ eV}$$

Photoelectrons with kE_{max} are moving on circular path.

$$r = \frac{mv}{qB}$$

$$mv = qBr$$

$$P = qBr = 1.6 \times 10^{-19} \times \frac{1}{3200} \times 10^{-3}$$

$$\frac{1}{2} \times 10^{-24} = 5 \times 10^{-25} \text{ kg m/s}$$

Energy of photoelectron = $KE_{max} = \frac{P^2}{2m}$

$$= \frac{25 \times 10^{-50}}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= 0.86 \text{ eV}$$

Now use Einstein equation

$$h\nu = \phi + kE_{max}$$

$$1.89 = 0.56 + \phi \; ; \; \phi = 1.03 \text{ ev}$$

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21.

$$\left(P_{avg}\right)_{left} hL = \left(P_{avg}\right)_{right} RL$$

$$\Rightarrow \frac{1}{2}(2\rho gh)(hL) = \frac{1}{2}(3\rho gR)(RL)$$

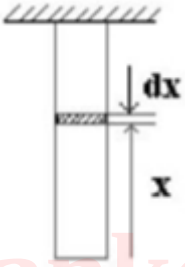
$$\Rightarrow h = \sqrt{\frac{3}{2}}R$$

22. $T(x) = \frac{mgx}{L}$

Elongation in element

$$= \frac{T\,dx}{YA} = \frac{1}{YA}\frac{mg}{L}x\,dx$$

$$\therefore \text{ Net elongation} = \frac{mgL}{2YA}$$



23. $\sin 30^0 = \frac{\lambda}{a}$

$$\Rightarrow \sin 30^o = \frac{6500}{a}\text{ Å}$$

$$\therefore \frac{1}{2} = \frac{6500}{a}\text{ Å}$$

$$\therefore \quad a = 2 \times 6500\text{ Å} = 2 \times 6500 \times 10^{-10}\text{ m}$$

$$\therefore \quad a = 1.3 \times 10^{-6}\text{ m}$$

24. $V = \sqrt{\gamma P/\rho}$

$$330 = \sqrt{\gamma \times \frac{1\times10^5}{1.3}}$$

$$\frac{(33)^2 \times 100 \times 1.3}{1\times10^5} = \gamma$$

$$\frac{1.089\times10^3\times10^2\times1.3}{1\times10^5} = \gamma$$

$$\frac{2}{f} + 1 = \gamma = 1.4 = 7/5$$

$$\frac{2}{f} = 2/5$$

$$f = 5$$

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25. $v_{avg} = \frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2}$

$= \frac{2\times3(4.5+7.5)}{6+4.5+7.5}\text{m/s}$

$= \frac{6\times12}{18}\text{m/s} = 4\text{m/s}$

CHEMISTRY

26. $\rightarrow a,c,d,e,h \rightarrow$ follows $(4n+2)\pi e^-$ Huckel's rule aromatic

$\rightarrow b,f,g \rightarrow$ follows $(4n)\pi e^-$ Huckel's rule anti aromatic

27. Addition takes place according to stability of carbocation

28. The order of first four elements of group-17 are as follows

$\rightarrow F > Cl > Br > I$ (electronegatively)

$\rightarrow Cl_2 > Br_2 > F_2 > I_2$ (Bond dissociation enthalpy in kj/mole)

$\rightarrow F_2 < Cl_2 < Br_2 < I_2$ (Melting point in K)

$\rightarrow F_2 < Cl_2 < Br < I_2$ (Boiling point in K)

29. According to Henry's law:

$P = k_H S$

Or

$S = \frac{1}{k_H} P$

So, for an **S vs P graph**:

Slope = $\frac{1}{k_H}$

Larger slope ⇒ smaller Henry's constant

30. **Statement I — Correct**

From Bohr's model:

$$v_n = \frac{Ze^2}{2\varepsilon_0 h}\cdot\frac{1}{n}$$

So, for a given Z, electron speed **decreases as n increases**.

Statement II — Correct

In quantum mechanics:

Electrons do **not move in definite circular orbits**

Hence terms like *orbital speed* are avoided

We describe electrons using **orbitals and probability distributions**

(Explicitly stated in NCERT)

Statement III — Correct

Frequency of emitted radiation:

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$$\nu = RZ^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$$

When taking **ratio of frequencies**, Z^2 cancels out.
Thus, ratio is **independent of** Z.
Statement IV — Correct
Blackbody radiation → Planck's quantum hypothesis

Photoelectric effect → Einstein's photon theory

Both require energy exchange in **discrete packets (quanta)**.

31. For free expansion of gas $P_{ext}=0$

$\therefore w=0, q=0 and\ \Delta T=0$

32. $\chi_{Benzene}=\dfrac{1}{(3/2)}=\dfrac{2}{3}$

$\therefore\ P_{Benzene}=P^0_{Benzene}\cdot\chi_{Benzene}$

$=75\times\dfrac{2}{3}=50\,\text{torr}$

33. moles of $Cu_2O=\dfrac{2.86\times10^3}{143}=20$

moles of $Cu_2S=\dfrac{4.77\times10^3}{159}=30$

Cu_2O is limiting reagent

$2Cu_2O+$	$Cu_2S\rightarrow$	$6Cu+$	SO_2
20	30	–	–
–	10	60	20

Mass of copper $=60\times63.5=3810g$

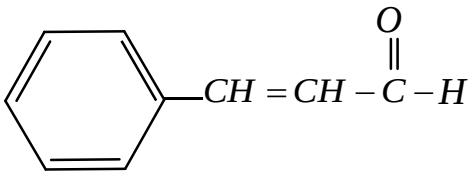
34. $K_{sp}<$ ionic product

35.

Molecule	Hybridization	Dipole moment
BF_3	Sp^2	Zero
$CHCl_3$	Sp^3	non-zero
CO_2	Sp	Zero
PCl_5	Sp^3d	Zero
XeF_4	Sp^3d^2	Zero
XeF_2	Sp^3d	Zero
CCl_4	Sp^3	Zero

36. Cinnamaldehyde

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is aliphatic aldehyde and therefore it gives positive test with tollens and fehling's reagent

→aromatic aldehyde will not give positive test with fehling's reagent

37. Glycine is optically inactive but valine, alanine ……are optically active

→ Isoleucine has 2 chiral centers

→Glutamic acid also containe COOH group at the side chain

→ Cysteine and methionine has sulphur containing amino acids

38. Basic strength of amine depends on hydrogen bonding and inductive effect

$\rightarrow \therefore (CH_3)_2 NH > CH_3 - NH_2 > (CH_3)_3 N > NH_3$

39. (d) For chlorine, the atomic radius and covalent radius are taken as same. Hence, statement I is incorrect. In case of anionic species, extra electrons are added in the parent atom, so effective nuclear charge decrease. Hence, anionic radius always greater than their parent atomic radius.

40. **Assertion (A): TRUE**

Oxidation state:

$$Fe^{3+} \Rightarrow d^5$$

CN^- is a **strong-field ligand** → low-spin octahedral complex

Electronic configuration:

$$t_{2g}^5 e_g^0$$

All electrons are in t_{2g}, **but one remains unpaired**

Hence the complex is **paramagnetic**

41. A) Prussian blue- $Fe_4\left[Fe(CN)_6\right]_3$

B) Violet-$\left[Fe(CN)_5 NOS\right]^{4-}$

C) Blood red-$\left[Fe(SCN)\right]^{2+}$

D) White- AgCl

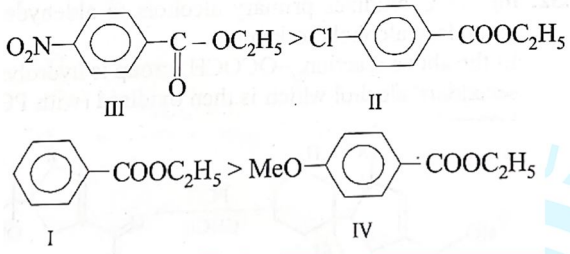
42. $\%C = \frac{12}{44} \times \frac{\text{weight of } CO_2}{\text{weight of organic compound}} \times 100$

$\% = \frac{2}{18} \times \frac{\text{weight of } H_2O}{\text{weight of organic compound}} \times 100$

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43. Rate of abstraction of hydrogen α stability of carbon free radical

44. Rate of reaction $\propto$ Electrophilicity of carbonyl carbons, so E.W.G increases the rate, while E.R.G decreases the rate



45. (i) In XeF_5^-, Steric No. =Bond pair+Lone Pair

$=(5+2)=7$

So, hybridisation is $=sp^3d^3$

and structure is pentagonal planar.

(ii) XeO_3F_2 Steric no=5

So, hybridisation is = sp^3 d

and structure is trigonal bipyramidal

46. In case of conversion of $Cr^{3+} \rightarrow Cr^{2}+$ the e^- is added to e_g orbital, hence electrode potential is negative. $E^0_{Cr^{+3}/Cr^{+2}} = -0.41V; \left[Cr^{+3}\right] = 4s^0 3d^3$

$$\mu = \sqrt{n(n+2)}B.M = \sqrt{15}B.M \sim 4B.M$$

47. $MnO_4^-(0.1M)/Mn^{2+}(0.001M)$

Reaction at cathode : $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

$$E = E^0 - \frac{0.059}{5}\log\frac{\left[Mn^{2+}\right]}{\left[MnO_4^-\right]\left[H^+\right]^8}$$

$$1.282 = 1.54 - \frac{0.059}{2}\log\frac{10^{-3}}{10^{-1}\times\left[H^+\right]^8}$$

$$\frac{0.258\times 5}{0.059} = \log\frac{10^{-2}}{\left[H^+\right]^8}$$

$$21.86 = -2 + 8pH$$

$$pH = 2.98 \approx 3$$

50. Statement evaluations:

1) True — linear 1/[A] vs t ⇒ second order.

2) True — matches the integrated form for second order.

3) False — incorrect units; for second order units are $L\cdot mol^{-1}\cdot s^{-1}$ (so this

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statement is wrong).

4) True — initial rate = $k[A]_0^2 = 0.020 \times (2.00)^2 = 0.080$ mol L^{-1} s^{-1}.

5) True — $t_{1/2} = 1/(k[A]_0) = 1/(0.020 \times 2.00) = 25.0$ s.

6) True — set $1/[A] = kt + 1/[A]_0$ with $[A] = 0.50$: $\frac{1}{0.50} = 2.00 = 0.020t + 0.50 \Rightarrow t = \frac{2.00-0.50}{0.020} = 75.0$ s.

7) False — for second order $t_{1/2} = 1/(k[A]_0)$, so doubling $[A]_0$ halves $t_{1/2}$, not doubles it.

Thus statements **3** and **7** are incorrect → **2 incorrect statements**.

MATHEMATICS

51. $\left|z_1+z_2\right|^2+\left|z_2+z_3\right|^2+\left|z_3+z_1\right|^2 \geq 12$

52.

$$(\sin x\cos y)\left(f(x+y)-f(x-y)\right)=(\cos x\sin y)\left(f(x+y)+f(x-y)\right)$$

$$f(x+y)\left(\sin(x-y)\right)=f(x-y)\sin(x+y)$$

$$\frac{f(x+y)}{\sin(x+y)}=\frac{f(x-y)}{\sin(x-y)}$$

Put $x+y=m, x-y=n$

$$\frac{f(m)}{\sin(m)}=\frac{f(n)}{\sin(n)}=K$$

$$\Rightarrow f(m)=K\sin(m)$$

$$\therefore f(x)=K\sin(x)$$

$$f'(x)=K\cos(x)$$

$$k=1$$

$$f'(\alpha)=\cos\alpha$$

$$f''(\alpha)=-\sin\alpha$$

$$f''\left(\frac{5\pi}{3}\right)=-\sin\left(\frac{\pi}{3}\right)$$

$$=\frac{-1}{2}$$

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53. $(SP)(S'P) = a(1-e\cos\theta)a(1+e\cos\theta)$

$= a^2 - a^2e^2\cos^2\theta$

$= 25 - 9\cos^2\theta$

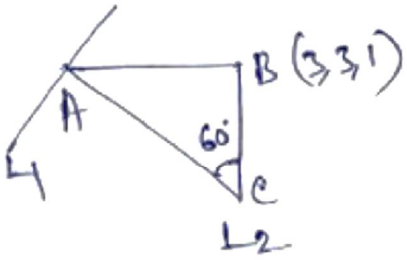
$\max = 25 - 9(0) = 25$

$\min = 25 - 9(1) = 16$

$\max - \min = 25 - 16 = 9$

54. Intersection point $A \equiv (1,1,1)$

Let C be $(1, 1+\mu, 1+\mu)$



$\overline{AC} = (0,1,1), \overline{BC} = (2, 2-\mu, -\mu)$

$\Rightarrow \cos 60 = \dfrac{0+2-\mu-\mu}{\sqrt{2}\sqrt{4+(2-\mu)^2+\mu^2}}$

$\Rightarrow (4+4\mu)^2 = 2\left(4+4+\mu^2-4\mu+\mu^2\right)$

$\Rightarrow 12\mu^2 - 24\mu = 0$

$$\Delta = \left|\frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0 & 2 \\ -2 & -2 & 0 \end{vmatrix}\right| = \left|\frac{1}{2}\left(4\hat{i} - 4\hat{j} + 4\hat{k}\right)\right|$$

$= \left|2\hat{i} - 2\hat{j} + 2\hat{k}\right|$

55. $\alpha^{15} + \beta^{15} = 1364$

$\alpha^{14} + \beta^{14} = 843$

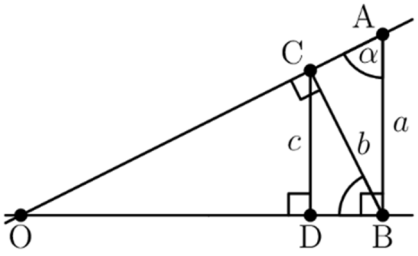
$\alpha^{13} + \beta^{13} = 521$

$P_{15} = P_{14} + P_{13}$

$x^2 = x + 1 \Rightarrow x^2 - x - 1 = 0$

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56.



$\angle OAB = \alpha$

so $\angle CBO = \alpha$

In ΔABC, we see that $\sin\alpha = \frac{b}{a}$ and in ΔCDB

$\sin\alpha = \frac{c}{b}$

$b = \sqrt{ac}$

Thus, because the length of a given perpendicular (except the first one) is the geometric mean of the two adjacent perpendiculars, the lengths of the perpendiculars

form a geometric sequence.

Because the sequence`s first two terms are a and b, it has common ratio $\frac{b}{a}$.

This infinite sum is given by $\frac{a}{1-\frac{b}{a}} = \frac{a^2}{a-b}$.

57. $b = \frac{-2}{3}$

58. A point on $x + y = 1$ *is* $(\alpha, 1-\alpha)$

The chord with mid-point $(\alpha, 1-\alpha)$ is

$(1-\alpha)y - 2a(x+\alpha) = (1-\alpha)^2 - 4a\alpha$

It passes through $(a, 2a)$.

$\therefore (1-\alpha)^2 - 4a\alpha = 2(1-\alpha)a - 2a(a-\alpha)$

$\Rightarrow \quad (1-\alpha)^2 = 2a(1-a)$

α has two distinct value so.

$2a(1-a) = (1-\alpha)^2 > 0 \Rightarrow 0 < a < 1\ i,e, 0 < 4a < 4$

$\therefore$ Length of latus rectum can be 1,2,3.

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59. Note that the store has the shape of a trapezium. So the cross-sectional area (A) of the store is given by the formula:

Area = average length of parallel sides x distance between parallel sides:

$$A = \frac{1}{2}(y+3)x$$

The lengths x and y are related through the fact that AB + BC = 6, where BC = y and

$AB = \sqrt{x^2 + (3-y)^2}$. Hence $\sqrt{x^2 + (3-y)^2} + y = 6$

which implies that $x^2 + 6y = 27$ .....(2)

$$A = \frac{1}{2}\left(\frac{27 - x^2}{6} + 3\right)x = \frac{1}{2}\left(\frac{27 - x^2 + 18}{6}\right)x = \frac{1}{12}(45x - x^3)$$

60. $$1 = \left|(\vec{b} - \vec{a}).\frac{(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right| \Rightarrow |\vec{b} - \vec{a}|\cos 60° = \frac{1}{2}AB \quad \Rightarrow \quad AB = 2$$

61. $f_n(x) = \tan^{-1}(x+n) - \tan^{-1}(x)\pi$

62. $$a = \int_0^{x_1} 1\,dx + \int_{x_1}^{1} 2\,dx$$

$$a = [x]_0^{x_1} + [2x]_{x_1}^1$$

$$a = (x_1 - 0) + (2(1) - 2(x_1))$$

$$a = x_1 + 2 - 2x_1$$

$$a = 2 - x_1$$

$$a = 2 - \sqrt{\frac{\sqrt{5}-1}{2}}$$

63. Maximum area of ABCD is 2ab

ABCD is a square, then its area is $\frac{4a^2b^2}{a^2+b^2}$

64. Let, $\sec x = t$

$$\lim_{t\to 1}\frac{t - t^t}{1 - t + \ln(t)} = \lim_{t\to 1}\frac{1 - t^t(1+\ln t)}{-1 + \frac{1}{t}} \text{ {by L-hospital rule}}$$

$$= \lim_{t\to 1}\frac{-t^t\left(\frac{1}{t}\right) - t^t(1+\ln t)^2}{-1/t^2} \text{ {again by L-hospital rule}}$$

$$= \frac{-1-1}{-1} = 2$$

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65. $R=\{f,g: f(2)=g(3) \text{ and } f(3)=g(2)\}$

Reflexive : (f,f) $\in R$

$= f(2)=f(3) \text{ and } f(3)=f(2) \rightarrow$ must hold

$\Rightarrow$ but this is not true for all function so not reflexive

Now, g(2)=f(3) and g(3)=f(2) $\rightarrow$ true

$\therefore$ symmetric

Transitive : If (f,g) $\in$ R and (g,h) $\in$ R

$\Rightarrow$ (f,h) $\in$ R

Now (f,g) $\in R \Rightarrow f(2)=g(3) \text{ and } f(3)=g(2)$

$(g,h)\in R \Rightarrow g(2)=h(3) \text{ and } g(3)=h(2)$

For $(f,h)\in R \Rightarrow g(2)=h(3) \text{ and } g(3)=h(2)$

Now $f(2)=g(3)=h(2) \text{ and } f(3)=g(2)=h(3)$

Hence not transitive

66. $\begin{vmatrix} 1 & -1 & 1 \\ 2 & 2 & \lambda_1 \\ 3 & (\lambda_1+4) & 1 \end{vmatrix}=0 \Rightarrow \lambda_1=1 \text{ or } -6 \text{ (no solution)}$

$\Rightarrow \lambda_1=1,\ \lambda_2=2,\ \lambda_3=4$

67. $\Rightarrow \vec{V}_2=x\vec{V}_1+y\hat{j}$

$\Rightarrow \vec{V}_2=x\left(2\hat{i}+2\hat{k}\right)+y\hat{j}$

$\therefore \left|\vec{V}_2\right|=\left|\vec{V}_1\right|=2\sqrt{2}$

$\therefore 4x^2+y^2+4x^2=8 \quad \ldots\ldots(1)$

Now. $\vec{V}_2.\vec{V}_1=\left|\vec{V}_2\right|\left|\vec{V}_1\right|\cos\frac{\pi}{3}=4$

$4=\left(x\left(2\hat{i}+2\hat{k}\right)+y\hat{j}\right).\left(2\hat{i}+2\hat{k}\right)$

$4=8x \Rightarrow x=\frac{1}{2} \Rightarrow y=\pm\sqrt{6}$

$\Rightarrow \vec{V}_2=\hat{i}+\sqrt{6}\hat{j}+\hat{k}\left(\because \vec{V}_2.\hat{j}>0\right)$

Now. $\vec{V}_3=\vec{V}_1+x\hat{i}$

$\vec{V}_3.\vec{V}_1=0 \Rightarrow 8+2x=0 \Rightarrow x=-4$

$\Rightarrow \vec{V}_3=-2\hat{i}+2\hat{k}$

$\Rightarrow$ Volume of tetrahedron

$V=\left|\frac{1}{6}\begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix}\right|$

$\Rightarrow V=\frac{1}{6}8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}}=4$

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68.

We have $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x\cos y}$

$\therefore \cos y\frac{dy}{dx} = \frac{\sin y + x}{2\sin y - x}$

Put $\sin y = vx$

$\therefore \cos y\frac{dy}{dx} = v + x\frac{dv}{dx}$

So, from (1), we have

$v + x\frac{dv}{dx} = \frac{vx + x}{2vx - x}$

$\Rightarrow x\frac{dv}{dx} = \frac{v+1}{2v-1} - v$

$\Rightarrow \int \frac{2v-1}{-2v^2+2v+1}dv = \int \frac{dx}{x}$

$\Rightarrow -\frac{1}{2}\log_e(-2v^2+2v+1) \log_e x + \log_e C$

$\Rightarrow \frac{\log_e 1}{\sqrt{-2v^2+2v+1}} = \log_e Cx$

$\Rightarrow \frac{1}{\sqrt{-2v^2+2v+1}} = Cx$

$\Rightarrow \frac{1}{\sqrt{-2\frac{\sin^2 y}{x^2} + 2\frac{\sin y}{x} + 1}} = Cx$

$\Rightarrow C_1 = -2\sin^2 y + 2x\sin y + x^2$

69.

$\left(4\tan x + \sec^2 x\right) = 2\sqrt{2}\sin\left(x + \frac{\pi}{4}\right)\sec^2 x$

$\Rightarrow 4\sin x\cos x + 1 = 2(\sin x + \cos x)$

$\sin x = \frac{1}{2}, \cos x = \frac{1}{2}$

$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{3}$

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70. $$S_1=\sum_{k=1}^{20}\binom{20}{k}\binom{20}{20-k+1}(-1)^k=\sum_{k=1}^{20}\binom{20}{k}\binom{20}{21-k}(-1)^k$$

This is the coefficient of x^{21} in the expansion of $(1-x)^{20}(1+x)^{20}=(1-x^2)^{20}$.

The general term in the expansion of $(1-x^2)^{20}$ is $\binom{20}{r}(-1)^r(x^2)^r=\binom{20}{r}(-1)^r x^2$

For the coefficient of x^{21}, we need $2r=21$, which has no integer solution for r.

Therefore, $S_1=0$.

$$S_2=\sum_{k=0}^{20}\binom{20}{k}\binom{20}{20-k}(-1)^k$$

This is the coefficient of x^{20} in the expansion of $(1-x)^{20}(1+x)^{20}=(1-x^2)$

For the coefficient of x^{20}, we need $2r=20$, so $r=10$.

The coefficient is $\binom{20}{10}(-1)^{10}=\binom{20}{10}$.

Therefore, $S_2=\binom{20}{10}$.

$$S_3=\sum_{k=0}^{21}\binom{21}{k}\binom{21}{21-k}(-1)^k$$

This is the coefficient of x^{21} in the expansion of $(1-x)^{21}(1+x)^{21}=(1-x^2)^{21}$.

The general term in the expansion of $(1-x^2)^{21}$ is $\binom{21}{r}(-1)^r(x^2)^r=\binom{21}{r}(-$

For the coefficient of x^{21}, we need $2r=21$, which has no integer solution for r.

Therefore, $S_3=0$.

71. We have $\int_0^x f(t)dt=x\sqrt{f(x)}$ .....(i)

Differentiating both the sides of equation (i) w.r.t. x, we get

$$f(x)=\frac{xf'(x)}{2\sqrt{f(x)}}+\sqrt{f(x)}$$

Let $f(x)=y^2$

$\Rightarrow$ $$f(x)=2y\frac{dy}{dx}$$

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APEX JEE-MAIN 2026
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$$y^2 = x.2y.\frac{dy}{dx}.\frac{1}{2y} + y$$

$$\Rightarrow \quad y^2 = x.\frac{dy}{dx} + y$$

$$\Rightarrow \quad y^2 - y = x.\frac{dy}{dx}$$

$$\int \frac{dy}{y(y-1)} = \int \frac{dx}{x}$$

$$\Rightarrow \quad \int \frac{y-(y-1)}{y(y-1)}dy = \int \frac{dx}{x}$$

$$\ln \frac{(y-1)}{y} = \ln cx$$

$$\Rightarrow \quad f(x) = \frac{1}{\left[1+\left(\sqrt{2}-1\right)x\right]^2}$$

$$\Rightarrow \quad f\left(\sqrt{2}+1\right) = \frac{1}{4}$$

72. Choosing (a, b) and (c, d) independently,

$$\text{Total outcomes} = 90^2 = 8100.$$

Let $N(s)$ be the number of ordered pairs $(a, b) \in S$ such that $a + b = s$.

For $s = 3, 4, \ldots, 19$, the values of $N(s)$ are:

s	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$N(s)$	2	2	4	4	6	6	8	8	10	8	8	6	6	4	4	2	2

If the sum is s, both (a, b) and (c, d) must lie among these $N(s)$ pairs, giving $N(s)^2$ favourable outcomes.

Hence,

$$\text{Favourable outcomes} = \sum_{s=3}^{19} N(s)^2.$$

Using symmetry,

$$\sum N(s)^2 = 2(2^2 + 2^2 + 4^2 + 4^2 + 6^2 + 6^2 + 8^2 + 8^2) + 10^2 = 580.$$

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73. Case 1: Digits $\{1,3,4\}$ in odd places (1st, 3rd, 5th) and $\{5,6,8\}$ in even plac 4th, 6th).

Number of arrangements = $3! \times 3! = 6 \times 6 = 36$.

Case 2: Digits $\{5,6,8\}$ in odd places and $\{1,3,4\}$ in even places.

Number of arrangements = $3! \times 3! = 6 \times 6 = 36$.

Total number of 6-digit numbers = $36 + 36 = 72$.

74. $r^2 = (\lambda+1)^2 + 1 = \left(\frac{|\lambda-2|}{\sqrt{2}}\right)^2$

$= \frac{(\lambda-2)^2}{2}$

$\lambda^2 + 8\lambda = 0$

$\Rightarrow \quad \lambda = , 0 - 8$

$\Rightarrow$ Equation of circle are

$x^2 + y^2 = 2$

and $(x+8)^2 + y^2 = 50$

$i.e. x^2 + y^2 - 2 = 0$

and $x^2 + y^2 + 16x + 14 = 0$

$r_1 + r_2 = \sqrt{2} + \sqrt{50} = 6\sqrt{2}$

75. Area $= \frac{8}{3} + 1 + \frac{1}{3} - \frac{5}{2} = \frac{3}{2}$

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